

JACC STATE-OF-THE-ART REVIEW

Aspirin Withdrawal in Patients With Coronary Artery Disease



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ABSTRACT

In patients with coronary artery disease (CAD), aspirin has long represented the cornerstone of antiplatelet therapy for secondary prevention. In patients with an acute coronary syndrome (ACS) and undergoing percutaneous coronary intervention (PCI), oral P2Y₁₂ inhibitors have traditionally been used as adjuncts to aspirin, a strategy known as dual antiplatelet therapy (DAPT). Although DAPT reduces the risk of ischemic recurrences, it carries a risk of bleeding. In CAD patients with a concomitant indication for oral anticoagulation (OAC)—who account for approximately 15% of patients—the combination of aspirin and OAC is associated with an increased risk of bleeding without ischemic benefit. The adverse impact of bleeding on prognosis has prompted investigations aimed at identifying antithrombotic treatment regimens associated with reduced bleeding without compromised ischemic protection. Among these, a strategy of aspirin withdrawal after a brief period of combination therapy has emerged as an attractive option. The efficacy and safety of aspirin withdrawal, however, vary substantially according to clinical setting, patient profile (eg, age, sex, ethnicity, genetic background, and comorbidities), timing of discontinuation after ACS or PCI, and alternative adjunctive antithrombotic therapy, including use of OAC and choice of P2Y₁₂ inhibitor. After reviewing the pharmacologic rationale for aspirin withdrawal, the clinical trial evidence, and guideline recommendations, we provide practical considerations for the implementation of this strategy in patients with CAD, with and without an indication to be on OAC. (JACC. 2026;88:1232-1254) © 2026 by the American College of Cardiology Foundation.

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ORAL ANTIPLATELET THERAPY IN CORONARY ARTERY DISEASE

Antiplatelet therapy is a cornerstone of secondary prevention in patients with coronary artery disease (CAD).¹⁻³ In particular, aspirin has long been the mainstay of antiplatelet agents, whereas in high-risk settings such as acute coronary syndrome (ACS) and percutaneous coronary interventions (PCI), oral P2Y₁₂ inhibitors (eg, clopidogrel, prasugrel, and ticagrelor) have traditionally been used as adjuncts to aspirin, a combination known as dual antiplatelet therapy (DAPT).^{1,4,5} The efficacy of DAPT in preventing both local (eg, stent thrombosis [ST]) and systemic (eg, recurrent myocardial infarction [MI]) thrombotic events has made DAPT the standard of care in ACS and PCI patients.¹ However, DAPT also increases the risk of bleeding.^{6,7} In patients with CAD requiring concomitant oral anticoagulation (OAC)—representing approximately 15% of the long-term secondary prevention population—the addition of aspirin to OAC increases the risk of bleeding without ischemic benefit.^{8,9} The ever-growing awareness of the adverse prognostic implications of bleeding, including increased mortality, has prompted investigations of antiplatelet treatment regimens aimed at mitigating the risk of bleeding without compromising efficacy.^{8,9}

Advances in PCI technology and procedural strategies (eg, intracoronary imaging), currently characterized by a low risk of local thrombotic complications, and the recognition that bleeding and thrombotic risks follow distinct temporal trajectories support time-adapted antiplatelet strategies.¹ Specifically, because bleeding risk increases with the intensity and duration of DAPT, whereas thrombotic risk is greatest within the first 1 to 3 months after the index event, strategies aimed at deescalating the intensity of platelet inhibition beyond this period have been investigated.^{1,10} Among these, withdrawal of aspirin therapy after a brief period of DAPT is associated with favorable outcomes.¹¹⁻¹⁵ However, the efficacy and safety of aspirin withdrawal may vary substantially according to clinical setting, patient profile (eg, age, sex, race, genetic background, and comorbidities), timing of discontinuation after ACS or PCI, and alternative adjunctive antithrombotic therapy, including use of OAC and choice of P2Y₁₂ inhibitor.¹⁶⁻²² In this review, we outline the pharmacologic rationale, critically appraise the available clinical trial evidence and its implications for practice guidelines, and provide practical recommendations to inform individualized decision making about aspirin discontinuation in

patients with CAD. Given that most of the evidence and corresponding guideline recommendations pertain to patients undergoing PCI, this review primarily focuses on that clinical setting.

PHARMACOLOGIC CONSIDERATIONS

Aspirin irreversibly inhibits the cyclooxygenase-1 enzyme, preventing thromboxane A₂-mediated platelet activation, whereas oral P2Y₁₂ inhibitors target the P2Y₁₂ receptor, a key amplifier of platelet signaling, and include thienopyridine and cyclopentyltriazolopyrimidine (CPTP) agents.^{5,23,24} Thienopyridines (clopidogrel and prasugrel) are prodrugs requiring hepatic bioactivation to generate an active metabolite that irreversibly inhibits P2Y₁₂, whereas the CPTP ticagrelor is direct-acting (ie, no metabolism required) and provides reversible inhibition.^{2,24-27} Clopidogrel, the most widely used P2Y₁₂ inhibitor, is characterized by marked interindividual variability in antiplatelet response.²⁸⁻³⁰ In particular, up to one-third of clopidogrel-treated patients exhibit high on-treatment platelet reactivity (HPR) and are at increased thrombotic risk, particularly after PCI.^{27,29,31,32} Clinical factors (eg, age, body mass index, chronic kidney disease, and diabetes) and genetic factors (eg, cytochrome P450 [CYP] 2C19 genotypes) contribute to clopidogrel response.³³ In particular, the CYP2C19 enzyme is involved in both metabolic steps required for clopidogrel conversion into its active metabolite, with carriers of CYP2C19 loss-of-function alleles having reduced metabolic activity.^{27,29,34} On the other hand, prasugrel has more efficient bioactivation that is not affected by CYP2C19 genotypes.^{26,35} Ticagrelor, instead, is not a prodrug and is also not modulated by CYP2C19 enzyme activity.³⁴ Both prasugrel and ticagrelor provide more potent and consistent P2Y₁₂ inhibition.²⁶ Although HPR with prasugrel and ticagrelor occurs in <5% of patients, the intense degree of platelet inhibition leads to low platelet reactivity in most patients, increasing the risk of bleeding.^{36,37} Indeed, growing evidence suggests that the superior efficacy of prasugrel and ticagrelor compared with clopidogrel is primarily driven by overcoming clopidogrel non-responsiveness, whereas among clopidogrel responders, prasugrel and ticagrelor may increase bleeding without ischemic benefit.^{36,38} These observations provide the rationale for platelet function testing (PFT) or CYP2C19 genotype testing to guide the selection of P2Y₁₂ inhibitors, favoring the use of

ABBREVIATIONS AND ACRONYMS

ACS	= acute coronary syndrome
AF	= atrial fibrillation
CAD	= coronary artery disease
CCS	= chronic coronary syndrome
DAPT	= dual antiplatelet therapy
OAC	= oral anticoagulation
PCI	= percutaneous coronary intervention

clopidogrel in responders and prasugrel or ticagrelor in nonresponders.²⁹ Importantly, the use of clopidogrel monotherapy (ie, without aspirin) raises concerns among clopidogrel nonresponders.^{9,29}

Pharmacodynamic studies have suggested that aspirin adds limited incremental platelet inhibition in the presence of potent P2Y₁₂ blockade, supporting clinical investigations using P2Y₁₂ inhibitor monotherapy with ticagrelor or prasugrel.³⁹ Ex vivo pharmacodynamic studies, primarily with ticagrelor, suggest that ticagrelor monotherapy and DAPT provide similar inhibition of adenosine diphosphate- and thrombin-induced platelet aggregation, whereas aspirin provides suppression of arachidonic acid- and collagen-mediated pathways.⁴⁰⁻⁴² However, aspirin and P2Y₁₂ inhibitors target distinct pathways, and the lack of a validated measure of global thrombogenicity complicates assessment of their individual effects.²⁹

EVOLVING PARADIGM IN ORAL ANTIPLATELET THERAPY

Aspirin was the first antiplatelet agent widely used for the treatment of patients with CAD. Pivotal randomized controlled trials (RCTs) conducted from the late 1970s through the late 1980s demonstrated that, compared with placebo, aspirin substantially reduced MI and cardiovascular (CV) death in patients with acute and chronic CAD.⁵ In 1996, despite clopidogrel showing better efficacy than aspirin in patients with atherosclerotic disease in a large RCT, the overall benefit was modest and not evident in the CAD subgroup.^{43,44} Accordingly, given its proven efficacy, low cost, and wide availability, aspirin has remained for decades the foundation of antiplatelet therapy in patients with CAD, and clopidogrel monotherapy was considered for patients who could not tolerate aspirin.^{4,5,44}

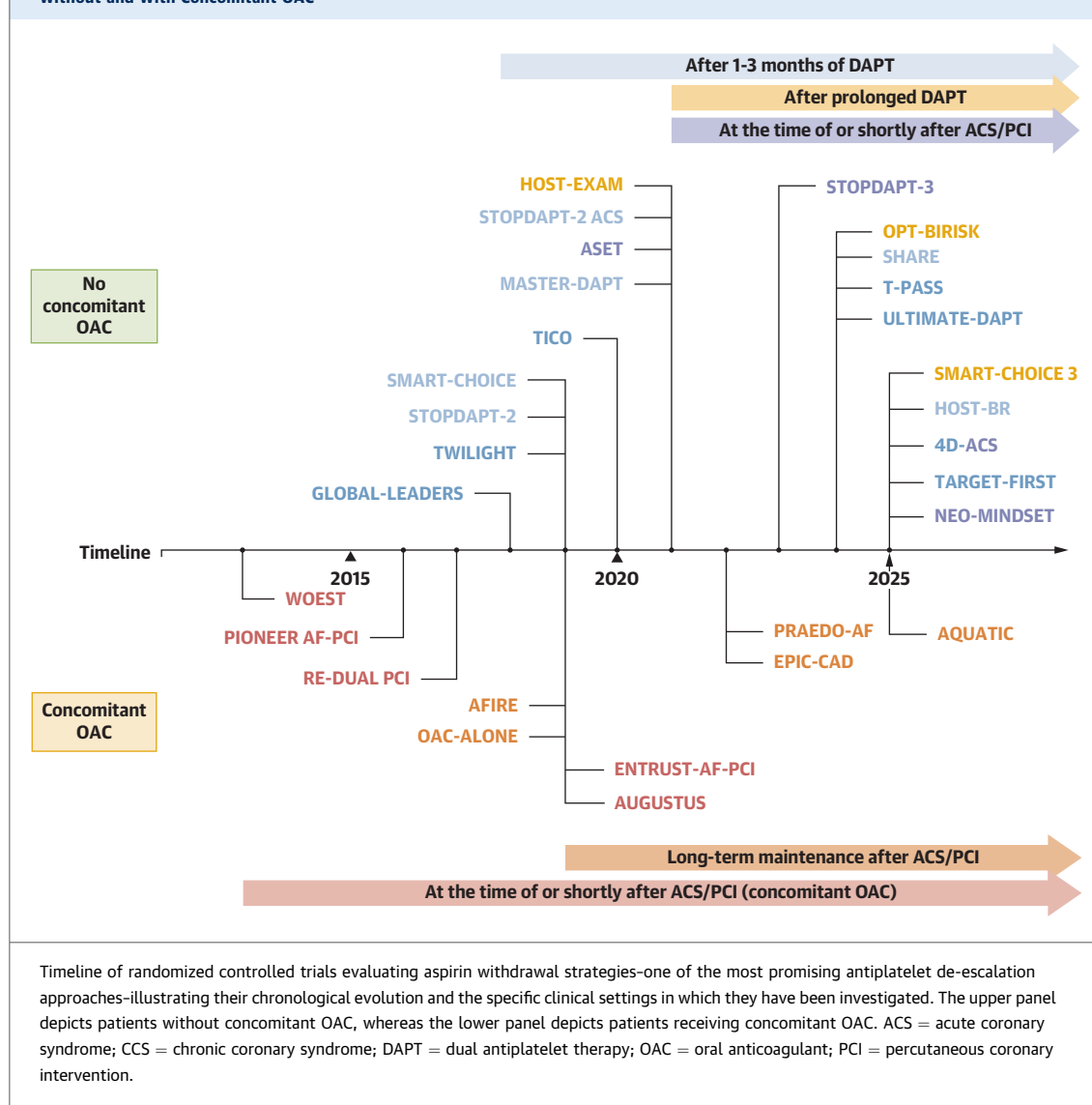
Between the late 1990s and early 2000s, RCTs demonstrated that, in patients with ACS undergoing PCI or not, adjunctive treatment with clopidogrel (ie, DAPT) was superior to aspirin alone and to alternative antithrombotic regimens in reducing ischemic events, including ST, while maintaining an overall acceptable safety profile, despite an increased bleeding risk compared with aspirin alone.^{29,45-49} As a result, DAPT became the antithrombotic strategy of choice in those high-risk settings.¹ During the first decade of the 2000s, pivotal RCTs showed that, compared with clopidogrel, post-ACS patients treated with aspirin plus prasugrel (only in PCI-treated patients) or aspirin plus ticagrelor (regardless of ACS management) experienced a greater reduction in ischemic recurrences, including ST,

albeit at the cost of increased major bleeding.^{50,51} Accordingly, in the absence of contraindications, DAPT with prasugrel or ticagrelor for at least 12 months has become the preferred treatment option for post-ACS patients.^{52,53} Although the trials were not designed to evaluate DAPT duration, a 12-month approach became the standard of care—particularly after ACS—based on the CURE trial (mean DAPT duration: ~9 months) and was subsequently reinforced by pivotal trials of potent P2Y₁₂ inhibitors.⁵⁴

The availability of 3 oral P2Y₁₂ inhibitors with different safety and efficacy profiles and the recognized contribution of CYP2C19 genotype and other factors on clopidogrel response prompted, in the early 2010s, studies of guided selection of P2Y₁₂ inhibitor therapy (eg, using PFT or genetic testing [GT]) to optimize the balance between bleeding and ischemic risk.^{29,55} However, earlier trials evaluating PFT-guided P2Y₁₂ inhibitor therapy failed to demonstrate a clinical benefit, likely owing to several factors, including limitations related to study design and availability of alternative agents.^{29,31,55} Subsequent studies and meta-analyses have shown more promising results, particularly when a guided selection of P2Y₁₂ inhibitor therapy was used as a strategy for de-escalation of intensity of P2Y₁₂ inhibition (eg, from prasugrel or ticagrelor to clopidogrel).⁵⁶⁻⁶⁰ Nevertheless, the limited availability of these tests, low familiarity with their use in clinical practice, and weak endorsement in current guidelines have prevented the routine implementation of guided P2Y₁₂ inhibitor selection.^{52,53,61,62}

Over the past decade, several developments have shifted attention from a “thrombosis era,” focused on intensifying or prolonging antithrombotic therapy to minimize ischemic events, to a “bleeding era,” in which the goal is to reduce bleeding without sacrificing antithrombotic protection (Figure 1).⁶³ This transition has been driven by increasing recognition that bleeding substantially worsens prognosis, including through excess bleeding-related mortality, the advent of contemporary stent platforms and PCI techniques (eg, intracoronary imaging-guided PCI, drug-coated balloons) associated with lower rates of thrombotic complications, including ST, and the recognition that thrombotic and bleeding risks vary widely among individuals and follow distinct temporal trajectories, with thrombotic risk peaking early (1-3 months) after ACS with or without PCI and declining thereafter, whereas bleeding risk is highest in the first week, decreases subsequently, but persists over time^{1,11-13,28,37,64} (Figure 1). In parallel, the availability of effective adjunctive therapies

FIGURE 1 Evolution of Antiplatelet Therapy and Timeline of Randomized Controlled Trials Evaluating Aspirin Withdrawal in Patients Without and With Concomitant OAC



targeting residual inflammatory, metabolic, and lipid risk providing synergistic protection and potentially exerting antiplatelet effects (eg, PCSK9 inhibitors, colchicine, icosapent ethyl)—has further supported a shift toward individualized and safer antithrombotic strategies, including de-escalation of antiplatelet therapy once the early post-ACS/PCI period of highest thrombotic risk has passed.^{1,3,11-13,64,65}

De-escalation antiplatelet strategies in ACS/PCI patients have included: 1) shorter DAPT duration followed by single antiplatelet therapy (SAPT); 2) switching from a more potent (prasugrel or ticagrelor) to a less potent (clopidogrel) P2Y₁₂ inhibitor, which can be guided (ie, with the use of PFT or GT) at the time of ACS/PCI or unguided after a brief course

(eg, 1 month) of standard DAPT; and 3) the use of reduced-dose P2Y₁₂ inhibition (eg, 5 mg prasugrel daily) after a short course of standard DAPT.^{1,11-13,64} Among these approaches, shortening DAPT duration has been the most intuitive and pragmatic strategy, and has therefore been investigated most extensively.^{64,66} Initial short-duration DAPT trials assessed stopping the P2Y₁₂ inhibitor after a brief course of DAPT and continuing aspirin monotherapy.^{1,11-13,64} However, in high thrombotic risk populations (eg, ACS), some studies suggest a potential trade-off between reduced bleeding risk and increased ischemic events compared with standard-duration DAPT.⁶⁷ Subsequently, studies have evaluated aspirin withdrawal after a short course of DAPT

with continuation of P2Y₁₂ inhibitor monotherapy.^{9,68} Overall, these trials have yielded encouraging findings, although outcomes may vary according to the specific P2Y₁₂ inhibitor used for monotherapy.^{11,64}

More recently, RCTs have tested the hypothesis of stopping aspirin early (ie, within 7 days) after PCI or ACS.⁶⁹⁻⁷¹ In parallel, other studies have suggested that clopidogrel may represent a preferable alternative to aspirin for long-term secondary prevention.⁴ Taken together, these data have fueled growing interest in aspirin-withdrawal strategies for secondary prevention in patients with CAD across a broad spectrum of clinical settings.

CLINICAL SETTINGS IN WHICH ASPIRIN WITHDRAWAL HAS BEEN TESTED

In this section, pivotal RCTs aimed at evaluating antiplatelet strategies for bleeding reduction through aspirin withdrawal to improve the balance between bleeding and ischemic risk in secondary prevention for patients with CAD are reviewed. These include withdrawal of aspirin therapy: 1) at the time of or shortly after PCI in ACS; 2) after 1 to 3 months of DAPT; 3) after prolonged DAPT; and 4) in patients receiving concomitant OAC (Tables 1 to 3, Figure 1).

Although aspirin withdrawal is expected to reduce bleeding compared with standard therapy, this must occur without any trade-off in efficacy. Therefore, interpretation of these trials should focus not only on safety but also on noninferiority regarding thrombotic events.⁹ Given the lower incidence of thrombotic than bleeding events, meta-analyses are often required to reliably assess effects on hard ischemic outcomes.^{9,72} In addition, the distinct ischemic bleeding profile of East Asian populations warrants caution when extrapolating findings to other racial groups.^{16,73-75} Notably, this occurs despite the higher prevalence of CYP2C19 loss-of-function alleles in Asian populations, which would be expected to increase thrombotic risk, particularly among clopidogrel-treated patients, a phenomenon commonly referred to as the “East Asian paradox.”^{73,74}

ASPIRIN WITHDRAWAL AT THE TIME OR SHORTLY AFTER PCI OR ACS. In 2020, 32 years after publication of the pivotal Second International Study of Infarct Survival (ISIS-2) trial, which established aspirin as a cornerstone therapy for the acute treatment of MI, the Acetyl Salicylic Elimination Trial (ASET) tested the hypothesis that aspirin could be discontinued at the time of PCI, maintaining prasugrel monotherapy, in elective post-ACS patients undergoing low-risk PCI.⁶⁹ A total of 201 patients were

enrolled in this single-arm open-label study, and at 3 months, the primary ischemic and bleeding endpoints occurred in 1 patient (0.5%) and no ST events were observed. These results, however, must be considered in the context of a small patient cohort with very low thrombotic risk.

The Short and Optimal Duration of Dual Antiplatelet Therapy-3 (STOPDAPT-3) trial, with enrollment ending in 2023, was a randomized open-label study that assigned 6,002 patients in Japan with ACS (75%, of whom 57% had ST-segment elevation myocardial infarction [STEMI]) or high bleeding risk (HBR) to either prasugrel monotherapy (3.75 mg/d) or DAPT with aspirin (81-100 mg/d) plus prasugrel (3.75 mg/d), after a 20-mg prasugrel loading dose administered in both groups immediately before PCI.⁷¹ At 1 month, the trial did not meet the primary endpoint of major bleeding (BARC 3 or 5) for superiority (4.47% in the treatment arm vs 4.71% in the comparator arm; $P = 0.66$), but met the co-primary endpoint of major adverse cardiovascular events (MACE) (composite of CV death, MI, definite ST, or ischemic stroke) for noninferiority with a relative 50% margin (4.12% vs 3.69%; $P_{\text{noninferiority}} = 0.01$).⁷¹ Key limitations of the study include the exclusive enrollment of East Asian patients, the use of a low-dose prasugrel regimen not approved in Western countries, the wide noninferiority margin, and the short follow-up.

In 2025, the Percutaneous Coronary Intervention Followed by Monotherapy Instead of Dual Antiplatelet Therapy in the Setting of Acute Coronary Syndromes (NEO-MINDSET) open-label trial tested whether early monotherapy with a potent P2Y₁₂ inhibitor (70% prasugrel, 30% ticagrelor), initiated shortly after successful PCI (mean 2 days), would be noninferior to DAPT with aspirin plus a potent P2Y₁₂ inhibitor (68% prasugrel, 32% ticagrelor) for preventing death or ischemic events (MI, stroke, urgent target vessel revascularization) and superior in reducing major or clinically relevant nonmajor bleeding in post-ACS patients (62% STEMI, 31% non-ST-segment elevation myocardial infarction [NSTEMI], 7% unstable angina [UA]) treated with contemporary drug-eluting stent (DES) platforms.⁷⁰ A total of 2,410 patients in Brazil were enrolled. At 12 months, potent P2Y₁₂ inhibitor monotherapy failed to show noninferiority to DAPT for the composite of death or ischemic events (7.0% vs 5.5%; $P_{\text{noninferiority}} = 0.11$). Notably, Kaplan-Meier curves diverged within the first month after ACS, and there were numerically higher rates of MI (2.7% vs 1.9%) and ST (0.7% vs 0.2%) in the monotherapy arm. The primary safety endpoint was not formally tested

TABLE 1 Randomized Controlled Trials Testing Aspirin Withdrawal in Secondary Prevention for Patients With Coronary Artery Disease

Study Name	Year of Publication and Trial Design	Region of Enrollment	No. of Patients Enrolled	Clinical Presentation, %		Treatment Arms and Study Population	Primary Endpoint Definition	Primary Endpoint Met?	Follow-Up Duration, mo
				ACS	CCS				
Aspirin withdrawal at the time or short after PCI/ACS									
ASET ⁶⁹	2020 (single-arm, open-label)	Brazil	201	0	100	Prasugrel monotherapy (aspirin discontinuation at the time of PCI) for 3 mo in patients treated with everolimus-eluting stent implantation with SYNTAX score <23	1) BARC 3 or 5 bleeding 2) MACE (CV death, spontaneous target vessel MI, or definite ST)	NA (single-arm)	3
STOPDAPT-3 ⁷¹	2023 (open-label)	Japan	6,002	75	25	Prasugrel 3.75 mg monotherapy vs 1-mo DAPT with aspirin and prasugrel 3.75 mg in patients with ACS or high bleeding risk just before PCI with cobalt-chromium everolimus-eluting stent	1) BARC 3 or 5 bleeding 2) MACE (CV death, MI, definite ST, or ischemic stroke)	No (bleeding: superiority) Yes (MACE: noninferiority)	1
NEO-MINDSET ⁷⁰	2025 (open-label)	Brazil	3,410	100	0	Potent P2Y ₁₂ inhibitor monotherapy (70% prasugrel, 30% ticagrelor) vs 12-mo standard DAPT (prasugrel 68%, ticagrelor 32%) in patients treated with successful PCI with DES	1) MACE (all-cause death, MI, stroke, or urgent target-vessel revascularization) 2) BARC 2, 3, or 5 bleeding	No (MACE: noninferiority) Not tested (bleeding: superiority)	12
Aspirin withdrawal after 1-3 months of DAPT									
GLOBAL-LEADERS ⁸³	2018 (open-label)	Multinational (18 countries)	15,968	47	53	Ticagrelor monotherapy for 23 mo vs DAPT with ticagrelor for 12 mo in patients treated with PCI with a Biolimus A9-eluting stent	All-cause death or MI	No (superiority)	24
SMART-CHOICE ⁷⁸	2019 (open-label)	South Korea	2,993	58	42	P2Y ₁₂ inhibitor monotherapy (clopidogrel 77%, prasugrel or ticagrelor 23%) after 3 mo of DAPT vs DAPT for 12 mo (clopidogrel 78%, prasugrel or ticagrelor 22%) in patients treated with PCI with DES	MACE (all-cause death, MI, or stroke)	Yes (noninferiority)	12
STOPDAPT-2 ⁷⁹	2019 (open-label)	Japan	3,045	38	62	P2Y ₁₂ monotherapy (clopidogrel 60%, prasugrel 3.75 mg 40%) after 1 mo of DAPT (clopidogrel 63%, prasugrel 3.75 mg 37%) vs 12-mo DAPT in patients treated with PCI with cobalt-chromium everolimus-eluting stent	NACE (CV death, MI, stroke, ST, or TIMI major or minor bleeding)	Yes (noninferiority and superiority)	12
TWILIGHT ¹³⁹	2019 (double-blind)	Multinational (11 countries)	7,119	64	36	Ticagrelor monotherapy after 3 mo of DAPT vs 12-mo ticagrelor-based DAPT in uneventful patients with high-risk PCI	BARC 2, 3, or 5 bleeding	Yes (superiority)	12
TICO ⁸⁶	2020 (open-label)	South Korea	3,056	100	0	Ticagrelor monotherapy after 3 mo of DAPT vs 12-mo ticagrelor-based DAPT in patients treated with DES	NACE (all-cause death, MI, ST, stroke, target-vessel revascularization, or TIMI major bleeding)	Yes (superiority)	12

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TABLE 1 Continued

Study Name	Year of Publication and Trial Design	Region of Enrollment	No. of Patients Enrolled	Clinical Presentation, %		Treatment Arms and Study Population	Primary Endpoint Definition	Primary Endpoint Met?	Follow-Up Duration, mo
				ACS	CCS				
STOPDAPT-2 ACS ⁸⁰	2022 (open-label)	Japan	4,169	100	0	P2Y ₁₂ inhibitor monotherapy (clopidogrel 52%, prasugrel 3.75 mg 48%) after 1-2 mo of DAPT vs 12-mo DAPT (clopidogrel 53%, prasugrel 3.75 mg 47%) among ACS in patients treated with PCI using cobalt-chromium everolimus-eluting stents	NACE (CV death, MI, stroke, ST, or TIMI major or minor bleeding)	No (noninferiority)	12
SHARE ⁸¹	2024 (open-label)	South Korea	1,387	74	26	3-mo DAPT followed by P2Y ₁₂ inhibitor monotherapy (clopidogrel 62%, ticagrelor 38%) vs 12-mo DAPT (clopidogrel 63%, ticagrelor 37%) in patients treated with PCI with DES	NACE (CV death, MI, ST, stroke, ischemia-driven target lesion revascularization, or BARC 3-5 bleeding)	Yes (noninferiority)	12
T-PASS ⁸⁷	2024 (open-label)	South Korea	2,850	100	0	Ticagrelor monotherapy after <1 mo of DAPT vs 12-mo ticagrelor-based DAPT in patients treated with PCI with DES	NACE (all-cause death, MI, definite or probable ST, stroke, or BARC 3-5 bleeding)	Yes (noninferiority and superiority)	12
ULTIMATE-DAPT ⁸⁸	2024 (double-blind)	Multinational (China, Pakistan, United Kingdom, and Italy)	3,400	100	0	1-mo DAPT followed by ticagrelor monotherapy vs 12-mo ticagrelor-based DAPT in patients who completed the IVUS-ACS ¹⁴⁰ study and who had no major ischemic or bleeding events after 1-mo treatment with DAPT	1) BARC 2-5 bleeding 2) MACE (CV death, MI, ischemic stroke, definite ST, or clinically driven target vessel revascularization)	Yes (bleeding: superiority) Yes (MACE: noninferiority)	12
4D-ACS ⁹⁰	2025 (open-label)	South Korea	656	100	0	1-mo DAPT with prasugrel 10 mg followed by prasugrel 5 mg monotherapy vs 12-mo DAPT with prasugrel 5 mg in patients treated with PCI with DES	NACE (all-cause death, MI, stroke, ischemia driven target vessel revascularization, or BARC 2, 3 or 5 bleeding)	Yes (noninferiority and superiority)	12
TARGET-FIRST ⁸⁹	2025 (open-label)	Europe	2,246	100	0	1-mo DAPT followed by P2Y ₁₂ inhibitor monotherapy (clopidogrel 4.8%, prasugrel 21.7%, ticagrelor 73.5%) vs 12-mo DAPT (clopidogrel 5.4%, prasugrel 20%, ticagrelor 74.6%) in patients with acute MI who had undergone revascularization within 7 days after MI	NACE (all-cause death, MI, ST, stroke, or BARC 3 or 5 bleeding)	Yes (noninferiority)	11
HOST-BR ⁸²	2025 (open-label)	South Korea	4,897	63	37	1- or 3-mo DAPT followed by P2Y ₁₂ monotherapy (mainly clopidogrel) vs 3- or 12-mo DAPT (mainly with clopidogrel) stratified according to HBR in patients treated with PCI with DES	1) NACE (all-cause death, MI, ST, stroke, or major bleeding) 2) MACE (cardiovascular death, MI, definite or probable ST, or ischemic stroke) 3) Any actionable nonsurgical bleeding	No in HBR, Yes in non-HBR (NACE: noninferiority) Yes in HBR (MACE: noninferiority) Yes in HBR (bleeding: superiority)	12

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TABLE 1 Continued

Study Name	Year of Publication and Trial Design	Region of Enrollment	No. of Patients Enrolled	Clinical Presentation, %		Treatment Arms and Study Population	Primary Endpoint Definition	Primary Endpoint Met?	Follow-Up Duration, mo
				ACS	CCS				
Aspirin withdrawal after prolonged DAPT									
HOST-EXAM ⁹⁶	2021 (open-label)	South Korea	5,438	72	28	Clopidogrel 75 mg vs aspirin 100 mg monotherapy in patients who maintained DAPT without clinical events for 6-18 mo after PCI with DES	NACE (all-cause death, nonfatal MI, stroke, readmission due to ACS, or BARC 3-5 bleeding)	Yes (superiority)	24
OPT-BIRISK ¹⁰⁴	2024 (double-blind)	China	7,758	100	0	Clopidogrel 75 mg vs DAPT with aspirin and clopidogrel in patients at increased bleeding and ischemic risk who had completed 9 mo of DAPT after ACS and remained free from adverse events for at least 6 mo before randomization	BARC 2, 3, or 5 bleeding	Yes (superiority)	9
SMART-CHOICE ³⁹⁹	2025 (open-label)	South Korea	5,506	24	76	Clopidogrel 75 mg vs aspirin 81-100 mg monotherapy in patients at high risk of recurrent ischemic events who completed a standard duration of DAPT after PCI	MACE (death from any cause, MI, or stroke)	Yes (superiority)	12
ACS = acute coronary syndrome; BARC = Bleeding Academic Research Consortium; CCS = chronic coronary syndrome; CV = cardiovascular; DAPT = dual antiplatelet therapy; DES = drug-eluting stent; HBR = high bleeding risk; MACE = major adverse cardiovascular events; MI = myocardial infarction; NA = not applicable; NACE = net adverse clinical events; PCI = percutaneous coronary intervention; ST = stent thrombosis; TIMI = Thrombolysis in Myocardial Infarction.									

owing to the ranked design, although bleeding was lower with monotherapy (2.0% vs 4.9%).⁷⁰ Subsequent secondary analyses of the trial suggested that aspirin withdrawal may be particularly detrimental during the first month after ACS or PCI and among patients presenting with STEMI.^{76,77} These findings continue to support that, nearly 40 years after ISIS-2, aspirin remains a fundamental component of therapy in high risk patients with ACS undergoing PCI, at least during the first month after ACS or PCI.

ASPIRIN WITHDRAWAL AFTER 1 TO 3 MONTHS OF DAPT. Discontinuation of aspirin after an initial period of DAPT has emerged as a promising strategy, but heterogeneity in outcomes across RCTs appears to depend on whether subsequent monotherapy consists of clopidogrel or a more potent P2Y₁₂ inhibitor.^{18,21}

Clopidogrel monotherapy. The open-label non-inferiority trial Comparison Between P2Y₁₂ Antagonist Monotherapy vs DAPT in Patients Undergoing Implantation of Coronary DES (SMART-CHOICE), conducted in South Korea, randomized 2,993 patients undergoing PCI with DES (58% ACS, 42% chronic coronary syndrome [CCS]) to P2Y₁₂ inhibitor monotherapy (clopidogrel 77%, prasugrel/ticagrelor 23%), after 3 months of DAPT vs standard 12-month

DAPT (clopidogrel 78%, prasugrel/ticagrelor 22%).⁷⁸ At 12 months, noninferiority for the primary endpoint of MACE (CV death, MI, or stroke) was demonstrated (2.9% vs 2.5%; $P_{\text{noninferiority}} = 0.007$). BARC 2-5 bleeding was significantly lower with P2Y₁₂ inhibitor monotherapy than with DAPT (2.0% vs 3.4%; $P = 0.02$).⁷⁸

The Short and Optimal Duration of Dual Antiplatelet Therapy-2 ACS Study (STOPDAPT-2) an open-label trial, randomized 3,045 patients in Japan undergoing PCI (38% ACS, 62% CCS) to P2Y₁₂ inhibitor monotherapy (clopidogrel 60%, 3.75 mg prasugrel 40%) after 1 month of DAPT or to 12-month DAPT (clopidogrel 63%, 3.75 mg prasugrel 37%).⁷⁹ At 12 months, P2Y₁₂ inhibitor monotherapy was both noninferior and superior for the primary endpoint of net adverse clinical events (NACE; CV death, MI, stroke, ST, or TIMI major/minor bleeding) (2.36% vs 3.70%; $P_{\text{noninferiority}} < 0.001$; $P_{\text{superiority}} = 0.04,81$). However, the STOPDAPT-2 ACS trial, which selectively enrolled 4,169 post-ACS patients (56% STEMI, 20% NSTEMI, 24% UA), failed to demonstrate non-inferiority for the primary endpoint of NACE with P2Y₁₂ inhibitor monotherapy (clopidogrel 52%, 3.75 mg prasugrel 48%) after 1-2 months of DAPT compared with 12-month DAPT (clopidogrel

TABLE 2 Randomized Controlled Trials Testing Aspirin Withdrawal in Secondary Prevention for Patients With Coronary Artery Disease Requiring Oral Anticoagulation

Study Name	Year of Publication and Trial Design	Region of Enrollment	No. of Patients Enrolled	Clinical Presentation, %		Treatment Arms and Study Population	Primary Endpoint Definition	Primary Endpoint Met?	Follow-Up Duration, mo
				ACS	CCS				
Early after ACS/PCI									
WOEST ¹¹¹	2013 (open-label)	Netherlands and Belgium	573	31	69	Clopidogrel +VKA (dual therapy) within a mean of 4 d since PCI vs clopidogrel + aspirin + VKA (triple therapy) in patients undergoing PCI with an indication for OAC	Any bleeding (defined by TIMI, GUSTO, and BARC)	Yes (superiority)	12
PIONEER AF-PCI ¹¹²	2016 (open-label)	Multinational	2,124	52	48	Rivaroxaban 15 mg + P2Y ₁₂ inhibitor therapy (clopidogrel 93.1%, prasugrel 1.7%, ticagrelor 5.2%) (group 1) within a mean of 3 d since PCI vs rivaroxaban (2.5 mg twice daily) + DAPT (clopidogrel 93.7%, prasugrel 1.6%, ticagrelor 4.8%) (group 2) vs VKA + DAPT (clopidogrel 96.3%, prasugrel 0.7%, ticagrelor 3%) (group 3) for a mean of 8 mo in patients with nonvalvular AF treated with PCI	TIMI major or minor bleeding or bleeding requiring medical attention	Yes (group 1 vs group 3; superiority)	12
RE-DUAL PCI ¹¹³	2017 (open-label)	Multinational (41 countries)	2,725	50	50	Warfarin + P2Y ₁₂ inhibitor (clopidogrel 90.3%, ticagrelor 7.8%) + aspirin for a mean of 2.7 mo (group 1) vs dabigatran 110 mg twice daily + P2Y ₁₂ inhibitor therapy (clopidogrel 86.4%, ticagrelor 12.6%) (group 2) within a mean of 5 d since PCI vs dabigatran 150 mg twice daily + P2Y ₁₂ inhibitor therapy (clopidogrel 86.9%, ticagrelor 12.1%) (group 3) within a mean of 5 d since PCI in patients with AF treated with PCI	ISTH major or clinically relevant nonmajor bleeding	Yes (noninferiority [group 1 vs group 2 + group 3]; superiority [group 1 vs group 2])	14
AUGUSTUS ³⁴	2019 (2 × 2: open-label VKA vs DOAC, double-blind aspirin vs placebo)	Multinational (33 countries)	4,614	38	62	Apixaban 5 mg or VKA + P2Y ₁₂ inhibitor therapy (clopidogrel 92.6%, prasugrel 1.1%, ticagrelor 6.2) within 6 d since PCI/ACS vs apixaban 5 mg or VKA + aspirin + P2Y ₁₂ inhibitor therapy for a mean of 6 mo in patients with AF after PCI or ACS	ISTH major or clinically relevant nonmajor bleeding	Yes (noninferiority and superiority)	6
ENTRUST-AF PCI ¹¹⁴	2019 (open-label)	Multinational (18 countries)	1,506	52	48	Edoxaban 60 mg + P2Y ₁₂ inhibitor therapy (clopidogrel 93%, prasugrel <1%, ticagrelor 7%) within 2 d since PCI vs VKA + P2Y ₁₂ inhibitor therapy (clopidogrel 92%, prasugrel <1%, ticagrelor 8%) + aspirin for a mean of 2 mo in patients with AF treated with PCI	ISTH major or clinically relevant nonmajor bleeding	Yes (noninferiority)	12

Continued on the next page

53%, 3.75 mg prasugrel 47%) (3.2% vs 2.8%; $P_{\text{noninferiority}} = 0.06$), largely driven by a higher incidence of MI in the monotherapy group (1.6% vs 0.9%).⁸⁰ The use of reduced-dose prasugrel and the selective inclusion of East Asian patients represent limitations of these trials.

SHARE (Short-Term Dual Antiplatelet Therapy After Deployment of Bioabsorbable Polymer

TABLE 2 Continued

Study Name	Year of Publication and Trial Design	Region of Enrollment	No. of Patients Enrolled	Clinical Presentation, %		Treatment Arms and Study Population	Primary Endpoint Definition	Primary Endpoint Met?	Follow-Up Duration, mo
				ACS	CCS				
Long term maintenance after ACS/PCI									
AFIRE ¹²⁷	2019 (open-label)	Japan	2,236	0	100	Rivaroxaban (15 mg; 10 mg if CrCl 15-49 mL/min) monotherapy vs rivaroxaban + SAPT (aspirin 69.9%, clopidogrel 24.9%, prasugrel 1.5%) in patients with AF and stable CAD	1) MACE (stroke, systemic embolism, MI, unstable angina requiring revascularization, or death from any cause) 2) ISTH major bleeding	Yes (noninferiority for MACE and superiority for major bleeding)	24
OAC-ALONE ¹²⁸	2019 (open-label)	Japan	696	0	100	OAC (VKA 75%, DOAC 25%) alone vs OAC + SAPT (aspirin or clopidogrel) in patients with AF and stable CAD	MACE (all-cause death, MI, stroke, or systemic embolism)	No (noninferiority; prematurely terminated)	30
PRAEDO-AF ¹²⁹	2022 (open-label)	Japan	147	0	100	Edoxaban 60 mg alone vs edoxaban 60 mg + clopidogrel in patients with AF and stable CAD	Composite of major bleeding or clinically significant bleeding (defined by ISTH)	No (superiority; underpowered)	20
EPIC-CAD ¹³⁰	2024 (open-label)	South Korea	1,040	0	100	Edoxaban 60 mg alone vs edoxaban + SAPT (aspirin 61.8%, clopidogrel 37.8%) in patients with AF and stable CAD	NACE (death from any cause, MI, stroke, systemic embolism, unplanned urgent revascularization, major bleeding, or clinically relevant nonmajor bleeding as defined by ISTH)	Yes (superiority)	12
AQUATIC ¹²⁶	2025 (double-blind)	France	872	0	100	OAC alone (DOAC 90%, VKA 10%) vs OAC + aspirin in patients with AF and stable CAD	MACE (CV death, MI, stroke, systemic embolism, coronary revascularization, or acute limb ischemia)	No (superiority)	26
ADAPT AF-DES ¹²⁵	2025 (open-label)	South Korea	960	0	100	DOAC alone (apixaban 61%, rivaroxaban 34%, edoxaban 4%, VKA 1%) vs DOAC + clopidogrel in patients with AF and stable CAD	NACE (death from any cause, MI, ST, stroke, systemic embolism, major bleeding, or clinically relevant nonmajor bleeding as defined by ISTH)	Yes (noninferiority)	12
ACS = acute coronary syndrome; AF = atrial fibrillation; CAD = coronary artery disease; CCS = chronic coronary syndrome; CrCl = creatinine clearance; CV = cardiovascular; DAPT = dual antiplatelet therapy; DOAC = direct oral anticoagulant; GUSTO = Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries criteria; ISTH = International Society on Thrombosis and Haemostasis; MACE = major adverse cardiovascular events; MI = myocardial infarction; NACE = net adverse clinical events; OAC = oral anticoagulation; PCI = percutaneous coronary intervention; SAPT = single antiplatelet therapy; ST = stent thrombosis; TIA = transient ischemic attack; TIMI = Thrombolysis in Myocardial Infarction; VKA = vitamin K antagonist.									

Everolimus-Eluting Stent) was an open-label non-inferiority trial randomizing 1387 South Korean patients who underwent PCI (74% ACS, 26% CCS) to 3-month DAPT followed by P2Y₁₂ inhibitor monotherapy (clopidogrel 62%, ticagrelor 38%) vs 12-month DAPT (clopidogrel 63%, ticagrelor 37%).⁸¹ Although the trial was limited by a lower than anticipated event rate (less than one-half of that projected for the primary endpoint), noninferiority was demonstrated for the primary outcome of NACE (defined as BARC 3 or 5 bleeding, CV death, MI, ST, stroke, or ischemia-driven target lesion revascularization) occurring between 3 and 12 months after the index PCI (1.7% vs 2.6%; $P_{\text{noninferiority}} < 0.001$).⁸¹

Among HBR patients undergoing PCI, the HOST-BR (Harmonizing Optimal Strategy for Treatment of Coronary Artery Diseases—DAPT Duration According to the Bleeding Risk) open-label trial compared shorter (1 month in HBR and 3 months in non-HBR) vs longer (3 months in HBR and 12 months in non-HBR) DAPT strategies according to bleeding risk in 3,299 South Korean patients (63% ACS, 37% CCS) undergoing PCI.⁸² After DAPT discontinuation, SAPT consisted of aspirin in 22.3%, clopidogrel in 68.2%, and a potent P2Y₁₂ inhibitor (prasugrel or ticagrelor) in 8.8% of patients.⁸² Three coprimary endpoints: 1) NACE (composite of all-cause death, MI, ST, stroke, or BARC 3 or 5 bleeding); 2) MACE (CV death, MI, ST, or ischemic stroke); and 3) BARC 2, 3, or 5

TABLE 3 Summary of Evidence of Aspirin Discontinuation Across Different Clinical Settings

Clinical Setting	Evidence Strength	Preferred Approach	Key Caveats
Aspirin withdrawal at time of or shortly after PCI/ACS Aspirin discontinuation <7 d after PCI for CCS or ACS	Limited by small sample size and use of reduced dose of P2Y ₁₂ inhibitors not approved for use	No indication	Evidence showed increased rates of thrombotic events in the aspirin-free group
Clopidogrel monotherapy after short course of DAPT Aspirin discontinuation and continuation of clopidogrel monotherapy after 1-3 mo of DAPT in patients undergoing PCI for CCS or ACS	Evidence limited by the use of noninferiority design, composite endpoints combining bleeding and ischemic events, and the nearly exclusive enrolment of East Asian patients. Impact of guided P2Y ₁₂ inhibitor selection in this setting not fully assessed.	Low ischemic risk (ie, CCS) or high bleeding risk patients	Increased risk of thrombotic events in high thrombotic risk (ie, ACS) patients
Ticagrelor monotherapy after short course of DAPT Aspirin discontinuation and continuation of ticagrelor monotherapy after 1-3 mo of DAPT in patients undergoing PCI for ACS	Solid evidence showing reduced risk of major bleeding by more than 50% and no trade-off in thrombotic events compared with standard ticagrelor-based DAPT. Possible enhanced effect in women compared with men.	ACS patients with uneventful course of 1-3 mo DAPT with ticagrelor	Though aspirin withdrawal after 3 mo appears safe regardless of clinical setting and ethnicity, the safety of discontinuation after only 1 mo remains uncertain in high-risk non-East Asian populations
Long-term aspirin vs clopidogrel monotherapy Use of clopidogrel in lieu of aspirin for long-term maintenance after DAPT interruption in patients with previous PCI	Data largely derived from East Asian populations	Clopidogrel may be used in place of aspirin in patients with previous PCI	Additional studies are warranted in non-East Asian patients
CAD plus OAC early after PCI Patients with either CCS or ACS undergoing PCI who required long-term OAC (most frequently for concomitant AF)	Individual trials not powered to definitively exclude increased major ischemic endpoints. Meta-analyses show possible increase in thrombotic events with short (<7 d) TAT compared to standard TAT in high risk patients (ie, ACS)	TAT duration may vary from 1 wk to 1 mo according to ischemic and bleeding risk	Concerns remain regarding aspirin withdrawal in patients with clopidogrel nonresponsiveness, as well as the role of more potent P2Y ₁₂ inhibitors
CAD plus OAC for long-term maintenance Patients with either CCS or ACS who have undergone PCI who require long-term OAC (most frequently for concomitant AF)	Individual trials limited by noninferiority design and modest statistical power for hard endpoints. Meta-analyses show significantly lower risks of any bleeding, major bleeding, CV death, and NACE, without evidence of increased composite or individual ischemic events compared with OAC plus SAPT	OAC monotherapy should be preferred over OAC plus SAPT in all patients with CCS requiring long-term OAC	It remains uncertain whether some subgroup of high ischemic risk patients may still derive a benefit from SAPT plus OAC

ACS = acute coronary syndrome; AF = atrial fibrillation; CCS = chronic coronary syndrome; CV = cardiovascular; DAPT = dual antiplatelet therapy; NACE = net adverse clinical events; PCI = percutaneous coronary intervention; OAC = oral anticoagulation; SAPT = single antiplatelet therapy; TAT = triple antithrombotic therapy (ie aspirin plus clopidogrel plus OAC).

bleeding—were assessed hierarchically at 1 year, with the first 2 tested for noninferiority and the third for superiority.⁸² In the HBR stratum, 1-month DAPT failed to meet noninferiority for NACE compared with 3-month DAPT (18.4% vs 14.0%; $P_{\text{noninferiority}} = 0.82$).⁸² MACE occurred numerically more frequently (9.8% vs 5.5%), whereas BARC 2, 3, or 5 bleeding occurred numerically less frequently (13.8% vs 15.8%), with 1-month DAPT.⁸² In contrast, in the non-HBR stratum, 3-month DAPT was non-inferior to 12-month DAPT for both NACE (2.9% vs 4.4%; $P_{\text{noninferiority}} < 0.0001$) and MACE (2.2% vs 2.3%; $P_{\text{noninferiority}} = 0.008$), and was superior in reducing BARC 2, 3, or 5 bleeding (7.4% vs 11.7%; $P < 0.0001$).⁸² Importantly, the findings underscore that the first 3 months after ACS or PCI represent a period

of heightened thrombotic risk regardless of bleeding risk status.¹ This has clinical relevance, because bleeding and thrombotic risks frequently overlap, complicating efforts to achieve an optimal balance between ischemic protection and bleeding avoidance.³⁷

Overall, the noninferiority design, use of composite endpoints combining bleeding and ischemic events—thereby limiting power for hard ischemic outcomes—the nearly exclusive enrollment of East Asian patients, and the signal toward higher event rates with clopidogrel monotherapy in trials selectively including patients with ACS undergoing PCI warrant caution regarding early aspirin discontinuation in high risk patients when clopidogrel is used (Table 3).

POTENT P2Y₁₂ INHIBITOR MONOTHERAPY. Published in 2018, GLOBAL-LEADERS: A Clinical Study Comparing Two Forms of Anti-platelet Therapy After Stent Implantation was the first and largest (open-label) trial to compare 1-month DAPT followed by 23 months of ticagrelor monotherapy vs 12-month ticagrelor-based DAPT followed by aspirin alone in 15,968 patients undergoing PCI (47% ACS, 53% CCS).⁸³ At 2 years, ticagrelor monotherapy after 1 month of DAPT was not superior for the prevention of death or MI (3.81% vs 4.37%; $P = 0.073$), and rates of BARC 3 or 5 bleeding were similar between groups (2.0% vs 2.1%).⁸³

The TWILIGHT (Ticagrelor With Aspirin or Alone in High Risk Patients After Coronary Intervention) trial was a double-blind placebo-controlled study that randomized 7,119 patients (64% ACS, 36% CCS) with at least 1 clinical and 1 angiographic feature conferring high ischemic or bleeding risk.⁸⁴ After 3 months of uneventful DAPT with ticagrelor plus aspirin, patients were assigned to ticagrelor with aspirin or placebo for 1 year.⁸⁴ Ticagrelor monotherapy significantly reduced the primary endpoint of BARC 2-5 bleeding by 44% (4.0% vs 7.1%; $P < 0.001$) without increasing MACE (3.9% vs 3.9%; $P_{\text{noninferiority}} < 0.001$) at 1 year.⁸⁴ Of note, the results of this trial, the only placebo-controlled double-blind study in this setting, were consistent in the pre-specified NSTEMI subgroup.⁸⁵

TICO (Ticagrelor Monotherapy After 3 Months in the Patients Treated With New-Generation Sirolimus-Eluting Stent for ACS), T-PASS (Ticagrelor Monotherapy in Patients Treated With New-Generation DES for ACS), and ULTIMATE-DAPT (Ticagrelor Monotherapy After 1 Month of DAPT Following PCI in Patients With an ACS) were 3 RCTs each enrolling nearly 3,000 East Asian patients with ACS who underwent short-duration DAPT (3 months in TICO, 1 month in T-PASS and ULTIMATE-DAPT), followed by ticagrelor monotherapy, compared with 12-month ticagrelor-based DAPT.⁸⁶⁻⁸⁸ TICO and T-PASS demonstrated superiority of short-duration DAPT for their primary endpoint of NACE (3.9% vs 5.9% [$P = 0.01$] and 2.8% vs 5.2 [$P = 0.002$], respectively).^{86,87} ULTIMATE-DAPT showed superiority of short-duration DAPT for BARC 2-5 bleeding (2.1% vs 4.6%; $P < 0.0001$), while meeting noninferiority for MACE (3.6% vs 3.7%; $P_{\text{noninferiority}} < 0.0001$) at 12 months.⁸⁸ Notably, ULTIMATE-DAPT used a double-blind placebo-controlled design and exclusively enrolled patients who underwent intravenous ultrasound-guided PCI and remained free from major ischemic or bleeding events during the first month of DAPT.⁸⁸

These findings were confirmed in European patients by the TARGET-FIRST (Evaluation of a Modified Anti-platelet Therapy Associated With Low-Dose DES Firehawk in Acute Myocardial Infarction Patients Treated With Complete Revascularization Strategy) trial.⁸⁹ Though open-label, this non-inferiority trial used NACE as the primary endpoint and enrolled 2,246 patients with ACS who underwent complete revascularization within 7 days of the index event (51% STEMI, 49% NSTEMI). Patients were randomized to 1-month DAPT followed by P2Y₁₂ inhibitor monotherapy (clopidogrel 4.8%, prasugrel 21.7%, ticagrelor 73.5%) or 12-month DAPT (clopidogrel 5.4%, prasugrel 20.0%, ticagrelor 74.6%).⁸⁹ At 11 months, 1-month DAPT followed by P2Y₁₂ inhibitor monotherapy was noninferior to standard DAPT for NACE (composite of all-cause death, MI, ST, stroke, or BARC 3 or 5 bleeding) (2.1% vs 2.2%; $P_{\text{noninferiority}} = 0.02$). Rates of ST (0.1% vs 0%) and BARC 3 or 5 bleeding (0.7% vs 0.7%) were low and similar between groups.⁸⁹

The open-label 4D-ACS (Effect of Short Duration of DAPT Followed by Dose Reduction After Implantation of DCS in ACS Patients) trial evaluated a combined de-escalation strategy consisting of early aspirin discontinuation and prasugrel dose reduction.⁹⁰ In 656 East Asian patients with ACS (33% STEMI, 31% NSTEMI, 36% UA), DAPT with aspirin and prasugrel (10 mg, or 5 mg in patients aged ≥ 75 years or weighing < 60 kg) was administered for 1 month, followed by aspirin withdrawal and prasugrel dose reduction to 5 mg daily.⁹⁰ At 12 months, 1-month DAPT was both noninferior and superior to standard DAPT for the primary endpoint of NACE (4.9% vs 8.8%; $P_{\text{noninferiority}} = 0.014$; $P_{\text{superiority}} = 0.034$), defined as a composite of death, MI, stroke, ischemia-driven target vessel revascularization, and BARC 2 to 5 bleeding.⁹⁰ The benefit was largely driven by a marked reduction in BARC 3 or 5 bleeding (0.6% vs 4.6%), whereas ischemic events were similar between groups (2.4% vs 3.7%).^{90,91} Despite its modest sample size, exclusive enrollment of East Asian patients, use of a noninferiority NACE endpoint, and evaluation of 2 concurrent de-escalation strategies (aspirin withdrawal and prasugrel dose reduction), this remains the only trial to date testing prasugrel monotherapy after short-duration DAPT.⁹¹

Overall, pooled analyses showed that ticagrelor monotherapy after short-duration (1-3 months) DAPT did not increase ischemic risk and reduced the risk of major bleeding by $> 50\%$ compared with standard ticagrelor-based DAPT, especially in post-ACS patients.⁹²⁻⁹⁴ Of note, there was an enhanced clinical

effect in women, despite their persistent underrepresentation in these trials.^{17,94,95}

ASPIRIN WITHDRAWAL AFTER PROLONGED DAPT.

In 1996, the pivotal CAPRIE (Clopidogrel Versus Aspirin in Patients at Risk of Ischaemic Events) trial compared 325 mg aspirin with 75 mg clopidogrel monotherapy in 19,185 patients with established atherosclerotic disease and showed a modest reduction in ischemic events with clopidogrel, driven mainly by patients with peripheral arterial disease, with no clear advantage over aspirin in those with recent (<35 days) MI.⁴³ Given aspirin's lower cost and widespread availability, it has remained the standard antiplatelet agent for secondary prevention in CAD.⁴ However, CAPRIE predates contemporary PCI practice, current adjunctive therapies, and modern aspirin dosing, and it included patients early after MI. Interest in clopidogrel as an alternative to aspirin has therefore re-emerged for long-term secondary prevention in patients with CAD or previous PCI.⁴

The open-label HOST-EXAM (Harmonizing Optimal Strategy for Treatment of Coronary Artery Stenosis—Extended Antiplatelet Monotherapy) trial randomized 5,530 South Korean patients who had remained event free on DAPT for 6 to 18 months after DES implantation to receive 75 mg clopidogrel or 100 mg aspirin monotherapy for 24 months.⁹⁶ At 2 years, clopidogrel significantly reduced the primary composite endpoint of all-cause death, MI, stroke, readmission for ACS, or BARC ≥ 3 bleeding compared with aspirin (5.7% vs 7.7%; $P = 0.003$). Importantly, clopidogrel also lowered the thrombotic composite endpoint (CV death, MI, ischemic stroke, ACS readmission, or ST; 3.7% vs 5.5%; $P = 0.003$), as well as ischemic stroke and BARC ≥ 2 and ≥ 3 bleeding. Although numerically higher rates of all-cause and CV death were observed with clopidogrel, the differences were not statistically significant and were no longer evident at extended follow-up through 5.8 years.^{97,98} Interpretation of these findings is tempered by the use of a composite endpoint incorporating softer outcomes as well as the exclusive enrollment of East Asian patients.⁷⁴

The open-label SMART-CHOICE 3 (Smart Angioplasty Research Team: Choice of Optimal Anti-thrombotic Strategy in Patients Undergoing Implantation of Coronary Drug-Eluting Stents 3) randomized 5,506 high risk patients who had completed standard DAPT after PCI to clopidogrel or aspirin monotherapy across 26 centers in South Korea.⁹⁹ High ischemic risk was defined by clinical (previous MI or diabetes) or procedural (complex PCI) features. Despite a lower than expected event rate, clopidogrel significantly reduced the primary endpoint of all-

cause death, MI, or stroke compared with aspirin (4.4% vs 6.6%; $P = 0.013$) over a median follow-up of 2.3 years. Reductions in death, MI, and ST numerically favored clopidogrel, whereas stroke and BARC 2-5 or 3-5 bleeding were similar between groups.⁹⁹ Interpretation is limited by the open-label design, underpowered primary endpoint, exclusive enrollment of East Asian patients, imbalance in all-cause mortality, and delayed separation of Kaplan-Meier curves, which emerged only after the second year, when fewer than one-half of patients remained under follow-up.¹⁰⁰

In the PANTHER individual participant data (IPD) meta-analysis (P2Y12 Inhibitor or Aspirin Monotherapy as Secondary Prevention in Patients With Coronary Artery Disease: An Individual Patient Data Meta-analysis of Randomized Trials), clopidogrel reduced the composite of CV death, MI, and stroke by 12% compared with aspirin overall, but a significant treatment-by-region interaction ($P_{\text{int}} = 0.034$) suggested that this benefit was largely confined to studies conducted in Asia.¹⁰¹ However, this analysis included patients treated with ticagrelor and those undergoing coronary artery bypass grafting, and it did not incorporate data from the more recent SMART-CHOICE 3 trial.¹⁰¹ A subsequent IPD meta-analysis focusing on post-PCI patients similarly reported a 14% reduction in CV death, MI, or stroke with clopidogrel vs aspirin, but did not identify a significant treatment-by-region interaction across Asia, Europe, and North America.¹⁰² A more recent meta-analysis found that, although the overall body of evidence suggests improved outcomes with clopidogrel compared with aspirin, a high level of confidence in these findings is currently limited to East Asian populations, and additional evidence is needed to clarify their applicability to non-East Asian patients.⁷⁵ Taken together, heterogeneity across trials and meta-analyses, coupled with ethnic differences in safety and efficacy outcomes associated with antiplatelet drugs, underscores the need for more studies across diverse patient populations to enhance our understanding of treatment with regimens with the most favorable outcomes.^{4,16,29,73,103}

OPT-BIRISK (Optimal Antiplatelet Therapy for High Bleeding and Ischemic Risk Patients Trial) was a double-blind placebo-controlled trial comparing 75 mg clopidogrel monotherapy with DAPT (aspirin plus clopidogrel) in 7,758 Chinese patients with ACS (21% STEMI, 25% NSTEMI, 54% UA) at increased bleeding and ischemic risk who had completed 9 to 12 months of DAPT after DES implantation and were event free for ≥ 6 months.¹⁰⁴ At 9 months, the primary endpoint of BARC 2, 3, or 5 bleeding was reduced with clopidogrel monotherapy (2.5% vs

3.3%; $P = 0.03$) compared with prolonged DAPT. Major adverse cardiac and cerebrovascular events (MACCE) (composite of all-cause death, MI, stroke, or clinically driven revascularization) was also reduced in the clopidogrel monotherapy arm (2.6% vs 3.5%; $P_{\text{noninferiority}} < 0.001$; $P_{\text{superiority}} = 0.02$), although BARC 3 or 5 bleeding was similar between groups (0.6% vs 0.7%).¹⁰⁴ This was the first trial to show the superiority of clopidogrel monotherapy over prolonged DAPT with aspirin and clopidogrel. However, the selective inclusion of event-free bi-risk patients exclusively enrolled in China, the lack of adjustment for multiplicity across secondary endpoints—including MACCE—and the absence of power for superiority analyses, together with similar rates of major bleeding, limit the generalizability of these findings, which should therefore be considered as hypothesis generating.

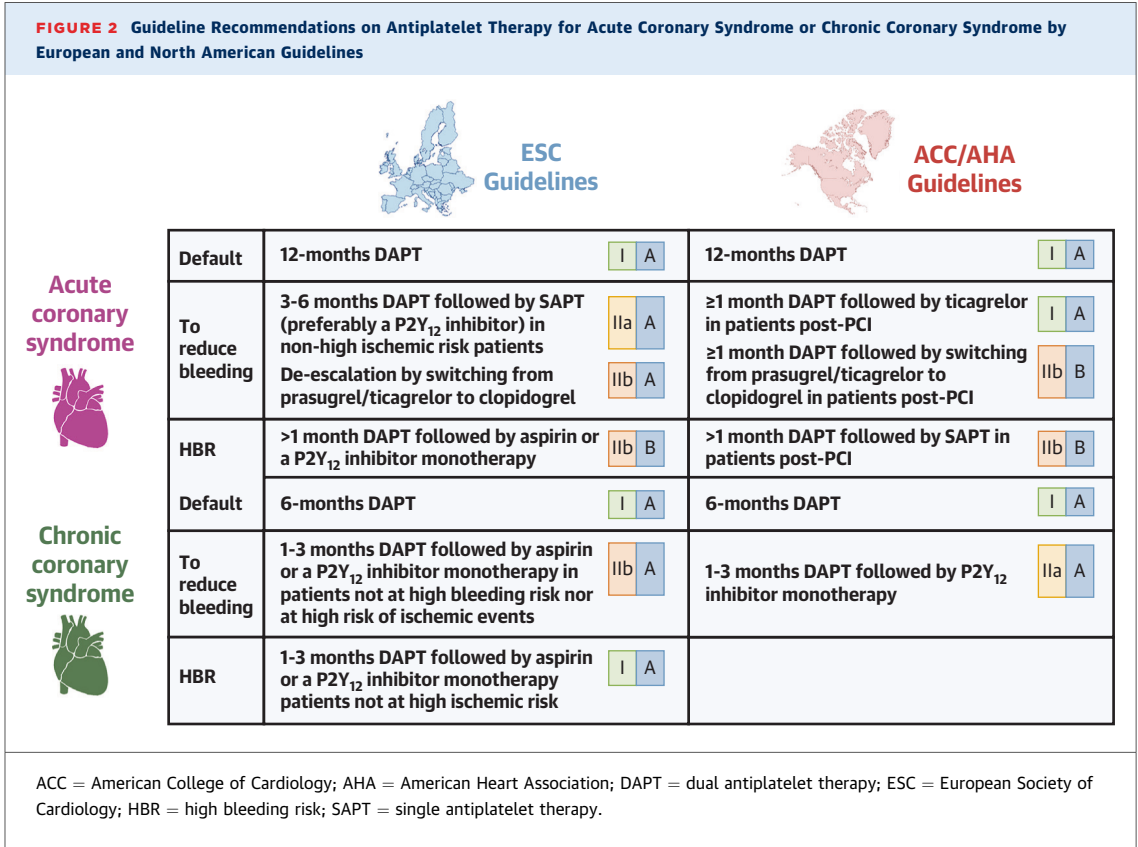
ASPIRIN WITHDRAWAL IN PATIENTS RECEIVING CONCOMITANT OAC. Atrial fibrillation (AF) and CAD frequently coexist, reflecting overlapping CV risk profiles and a shared tendency toward thrombogenesis.¹⁰⁵ Although antiplatelet therapy mitigates device-related (ie, ST) and arterial atherothrombotic complications, it provides inadequate protection against cardioembolic stroke in patients with AF or for secondary venous thromboembolism prevention in patients with previous venous thromboembolism, conditions for which OAC remains mandatory. Conversely, OAC alone does not sufficiently protect against arterial thrombotic events.¹⁰⁶ The addition of antiplatelet therapy to OAC is consistently associated with a pronounced increase in bleeding risk.^{6,7,106-110}

Patients with CAD receiving concomitant OAC represented the first clinical setting in which the safety and efficacy of an antiplatelet de-escalation strategy based on aspirin withdrawal were evaluated, with outcomes stratified according to the time elapsed since ACS or PCI.⁸ Because the present review does not primarily focus on patients treated with OAC, these 2 clinical scenarios are briefly summarized below (Table 2, Figure 1).

Early after ACS/PCI in combination with oral anticoagulation. Initial evidence from the landmark WOEST (What Is the Optimal Antiplatelet and Anticoagulant Therapy in Patients With Oral Anticoagulation And Coronary Stenting) trial showed that, in patients undergoing PCI who required long-term vitamin K antagonist (VKA) therapy, omission of aspirin with continuation of a P2Y₁₂ inhibitor reduced bleeding without an accompanying increase in thrombotic events.¹¹¹ This observation prompted a series of RCTs including patients with ACS and AF and evaluating double therapy with OAC and a P2Y₁₂

inhibitor (mostly clopidogrel) using all 4 currently available direct OACs—dabigatran, rivaroxaban, apixaban, and edoxaban.^{84,112-114} Collectively, these trials support the strategy of early aspirin discontinuation after a short course of DAPT (mean 4 days), with no significant excess of ischemic complications in most patients.⁶⁸ However, the optimal timing for aspirin withdrawal in these patients remains debated, with proposed windows ranging from hospital discharge or within the first week to 1 month after PCI.⁹ Nonetheless, for patients with AF and PCI/ACS there is broad consensus that continuation beyond 30 days is rarely justified.¹¹⁵⁻¹¹⁷ Importantly, individual trials were not sufficiently powered to definitively exclude increases in major ischemic endpoints, prompting multiple meta-analyses.⁶⁸ Though results have been heterogeneous, several analyses reported a higher incidence of thrombotic events with very early aspirin discontinuation, supporting maintenance of aspirin for at least 30 days in patients at elevated ischemic risk.^{72,118-122} Finally, because clopidogrel is the recommended antiplatelet agent in this context, concerns remain regarding aspirin withdrawal in patients with clopidogrel non-responsiveness, as well as the role of more potent P2Y₁₂ inhibitors—both of which warrant further investigation (Table 3).^{9,29,36,123,124}

Long-term maintenance after ACS/PCI. Long-term maintenance therapy after ACS/PCI generally refers to the 6-to-12-month period after PCI, depending on clinical presentation (ACS vs CCS) and thrombotic risk. Although growing evidence shows that thrombotic events are strongly time dependent—with up to 80% of ST occurring within the first month after ACS/PCI—and that contemporary stent platforms are associated with very low absolute rates of this rare but catastrophic complication, clinicians in real-world practice often remain reluctant to discontinue antiplatelet therapy and rely on OAC alone after PCI, especially for those with a history of complex PCI.^{109,125} Against this background, 6 RCTs reported over the past decade have consistently shown that omission of aspirin nearly halves bleeding risk, including major bleeding, without an apparent increase in thrombotic events.¹²⁵⁻¹³⁰ Of these, the recent AQUATIC (Assessment of Quitting Versus Using Aspirin Therapy in Patients With Stabilized Coronary Artery Disease After Stenting Who Require Long-Term Oral Anticoagulation) and ADAPT AF-DES (Appropriate Duration of Antiplatelet and Thrombotic Strategy After 12 Months in Patients With Atrial Fibrillation Treated With Drug-Eluting Stents) trials have provided additional evidence to this setting.^{125,126} Nevertheless, AQUATIC was



prematurely terminated after enrolling only 872 of the planned 2,000 patients owing to an excess of all-cause mortality in the SAPT plus OAC group, and ADAPT AF-DES was restricted to an East Asian population.^{125,126} All of these trials were limited by modest sample sizes, early termination, and the use of anticoagulation regimens no longer aligned with contemporary guidelines, including reduced-dose direct OACs (eg, 10-15 mg rivaroxaban daily) or VKAs,¹²⁵⁻¹³⁰ resulting in limited statistical power for hard endpoints, as well as by the use of a composite ischemia and bleeding primary endpoint, non-inferiority design, and lower than anticipated event rates. In this context, meta-analyses have helped to mitigate some of the limitations of individual trials, including limited statistical power, and consistently demonstrate that OAC monotherapy is associated with significantly lower risks of any bleeding, major bleeding, CV death, and NACE, without evidence of increased composite or individual ischemic events compared with OAC plus SAPT.¹³¹⁻¹³³

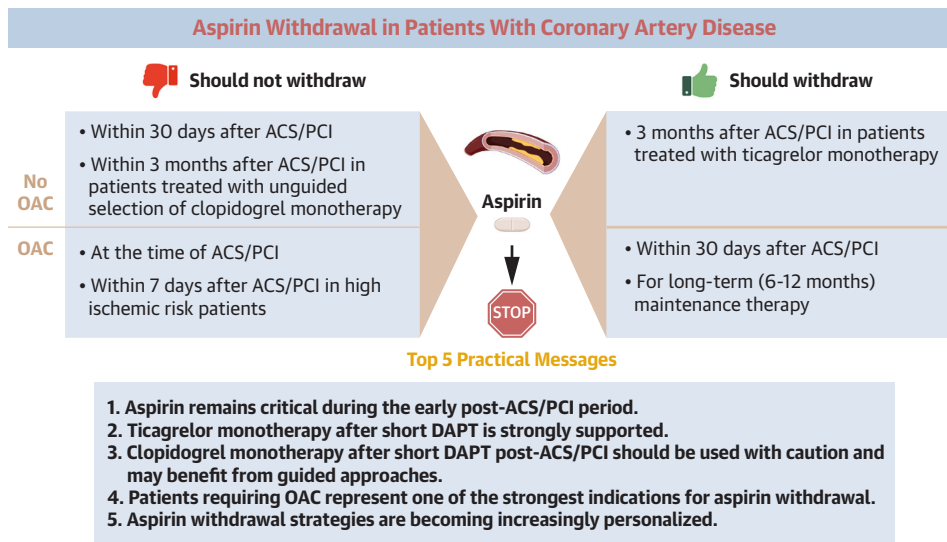
CURRENT GUIDELINE RECOMMENDATIONS

Despite the growing body of RCT evidence supporting antiplatelet deescalation strategies—particularly

aspirin withdrawal—both European and North American guidelines continue to recommend, as Class I, Level of Evidence (LoE) A, standard 12-month DAPT with prasugrel or ticagrelor for ACS and 6-month DAPT with clopidogrel for CCS.^{52,53,61,62} These recommendations have sparked substantial debate among experts.^{54,134} Notably, some differences exist between European and North American guidelines. An overview of these recommendations is provided in [Figure 2](#).

In ACS, North American guidelines—which are more recent than the European recommendations and hence with latest updates in the field (2025 vs 2023)—also endorse stopping aspirin and continuing ticagrelor monotherapy after PCI after at least 1 month of DAPT with ticagrelor (Class I, LoE A) to reduce bleeding.⁵³ In contrast, European guidelines provide a weaker and more general recommendation for SAPT after 3 to 6 months of DAPT, preferably with a P2Y₁₂ inhibitor (Class IIB, LoE A).⁵² Both guidelines acknowledge the option of an unguided de-escalation from ticagrelor to clopidogrel after completion of a short DAPT course.^{52,53} However, neither guideline addresses the use of guided de-escalation strategies, despite earlier guideline versions including specific recommendations.^{52,53,56-58,115} Finally, in patients

CENTRAL ILLUSTRATION Practical Recommendations of Aspirin Withdrawal for Secondary Prevention in Patients With Coronary Artery Disease



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ACS = acute coronary syndrome; DAPT = dual antiplatelet therapy; OAC = oral anticoagulant; PCI = percutaneous coronary intervention.

with HBR, both guideline documents allow SAPT with either aspirin or a P2Y₁₂ inhibitor after a minimum of 1 month of DAPT (Class IIb, LoE B).^{52,53}

In CCS, European guidelines are slightly more recent than the North American recommendations (2024 vs 2023). Both guidelines endorse the option of shortening DAPT duration to 1 to 3 months after PCI, but although European guidelines do not specify the SAPT agent to be used and recommend this approach to reduce bleeding in patients without HBR or high ischemic risk (Class IIb, LoE A; Class I, LoE A in HBR patients), North American guidelines specifically recommend continuation with a P2Y₁₂ inhibitor after DAPT (Class IIa, LoE A).^{61,62} Finally, both guidelines recommend aspirin for long-term secondary prevention in patients with CCS (Class I, LoE A). However, the European guidelines also endorse clopidogrel as a safe and effective alternative to aspirin monotherapy in patients with previous MI or PCI (Class I, LoE A).^{61,62}

BRIDGING EVIDENCE TO PRACTICE: PRACTICAL RECOMMENDATIONS

Aspirin withdrawal is associated with several potential advantages, including reduced burden of polypharmacy, dyspepsia, gastrointestinal ulcers, and bleeding, which may translate into improved adherence, reduced ischemic events, and improved

survival. Indeed, bleeding is closely associated with ischemic events—and may be causally related through direct effects (eg, hypotension, hypoxia), complications of blood transfusion or invasive interventions, and interruption of life-saving therapies, including antiplatelet agents, with an increase in platelet reactivity.^{7,135} However, in specific high risk settings, aspirin discontinuation may be associated with an increased risk of thrombotic events, potentially reflecting insufficient platelet inhibition in the context of heightened platelet reactivity and elevated thrombotic risk.

Based on current evidence, practical recommendations can be formulated to identify clinical scenarios in which aspirin withdrawal strategies may be appropriate or should be avoided in patients with CAD (Central Illustration). Among patients with an indication for OAC, there is substantial evidence to support aspirin withdrawal with continuation of clopidogrel, within 1 to 4 weeks after ACS/PCI, up to 30 days, but not beyond, in those at high ischemic and low bleeding risk. Antiplatelet therapy should be withdrawn for long-term (eg, 6-12 months) secondary prevention in those with concomitant indication for OAC. Among patients undergoing PCI, stopping aspirin and continuing ticagrelor monotherapy after 1 to 3 months after PCI, may be more widely implemented to reduce bleeding without compromising

TABLE 4 Ongoing Randomized Controlled Trials Testing Aspirin Withdrawal in Secondary Prevention for Patients With Coronary Artery Disease

Trial Name	Trial Design and Clinicaltrials.gov ID	No. of Patients to Be Enrolled	Population	Treatment Arm	Comparator Arm	Primary Endpoint(s)	Estimated Trial Completion Date
Without concomitant indication for OAC							
MATE	Open-label NCT04937699	2,690	ACS	1-mo DAPT with ticagrelor followed by ticagrelor monotherapy for 5 mo followed by clopidogrel monotherapy for 6 mo	12-mo ticagrelor-based DAPT	NACCE at 12 mo	December 2026
PROMOTE	Open-label NCT06916520	300	ACS or CCS	Prasugrel 5 mg daily monotherapy for 6 mo in CCS and for 12 mo in ACS	Standard 12-mo DAPT	NACE at 12 mo	April 2027
OPT-CAD	Open-label NCT06216821	3,490	ACS	P2Y ₁₂ inhibitor monotherapy after 1 mo of DAPT in low-risk patients and after 3 mo of DAPT in patients at moderate to high risk	Standard 12-mo DAPT	NACE at 12 mo	December 2027
STARS-DAPT	Open-label NCT05785897	350	STEMI	Prasugrel or ticagrelor monotherapy (stop aspirin at the time of PPCI)	Standard 12-mo DAPT	MACCE and BARC 3 or 5 bleeding at 1 y	March 2028
ADAPT-CKD	Open-label NCT04708587	900	PCI in CKD patients	DAPT for at least 3 mo followed by P2Y ₁₂ monotherapy	DAPT for at least 6 mo followed by P2Y ₁₂ monotherapy	NACE at 1 year	March 2028
PREMIUM	Open-label NCT05709626	2,258	STEMI	Prasugrel monotherapy for 12 mo	12-mo prasugrel-based DAPT	MACE at 1 year	April 2028
COMPARE STEMI-ONE	Open-label NCT05491200	1,656	STEMI	Prasugrel-based short DAPT (30–45 d) followed by prasugrel monotherapy	12-mo prasugrel-based DAPT	NACE at 11 mo	August 2028
ANGIODAPT	Open-label NCT05952206	2,312	ACS or CCS	1-mo DAPT with prasugrel or ticagrelor followed by the same P2Y ₁₂ inhibitor monotherapy in patients undergoing Angiolite or Xience stent implantation	Standard 6-mo (CCS) or 12-mo (ACS) DAPT in patients undergoing Angiolite or Xience stent implantation	BARC 2, 3, or 5 bleeding at 1 y	August 2028
MACT II	Open-label, single arm NCT06543082	490	ACS undergoing Orsiro DES implantation	Colchicine 0.6 mg daily + ticagrelor monotherapy (aspirin discontinuation the day after PCI)	–	NACE and ST at 12 mo	December 2028
OPTIMIZE-APT	Open-label NCT05418556	3,994	ACS or CCS	1-mo DAPT followed by clopidogrel monotherapy in CCS and 3-mo DAPT followed by ticagrelor monotherapy in ACS	Standard 12-mo DAPT	BARC 2, 3, or 5 bleeding, MACE, and NACE at 1 y	December 2028
TAILOR-DAPT	Open-label NCT05732701	2,788	ACS or CCS	1-mo DAPT followed by P2Y ₁₂ monotherapy (clopidogrel in CCS and ticagrelor in ACS) in HBR; 6–12-mo DAPT followed by P2Y ₁₂ monotherapy (clopidogrel in CCS and ticagrelor in ACS) in non-HBR according to PCI complexity	Standard 12-mo DAPT	NACE at 1 y	December 2028
Safety and Efficacy of De-escalation Dual Antiplatelet Therapy After BioFreedom Stenting in ACS Patients With Moderate-to-High Ischemic and High Bleeding Risk	Open-label NCT07374718	468	ACS with moderate-to-high ischemic risk and HBR	1-mo DAPT followed by 11-mo P2Y ₁₂ inhibitor monotherapy in patients undergoing IVUS-guided Biofreedom stent implantation	Standard 12-mo DAPT in patients undergoing angiography-guided DES	NACE at 12 mo	June 2029
SOLOPCI	Open-label NCT07164859	1,700	ACS or CCS	DAPT (prasugrel or ticagrelor in ACS and clopidogrel in CCS) for 7 d followed by P2Y ₁₂ inhibitor monotherapy	DAPT (prasugrel or ticagrelor in ACS and clopidogrel in CCS) for at least 3 mo	NACE at 12 mo	October 2029

Continued on the next page

TABLE 4 Continued

Trial Name	Trial Design and Clinicaltrials.gov ID	No. of Patients to Be Enrolled	Population	Treatment Arm	Comparator Arm	Primary Endpoint(s)	Estimated Trial Completion Date
GUIDE-HBR	Open-label NCT06763744	3,000	ACS with HBR	Genotype-guided P2Y ₁₂ monotherapy (clopidogrel in responder and prasugrel or ticagrelor in nonresponders) before discharge	Unguided deescalation from prasugrel/ticagrelor to clopidogrel after 1 mo of DAPT	BARC 2, 3, or 5 bleeding at 6 mo	December 2029
ONE-PASS	Open-label NCT05305482	3,520	ACS	1-mo DAPT with aspirin and ticagrelor followed by ticagrelor monotherapy after Biofreedom stent implantation	1-mo DAPT with aspirin and ticagrelor followed by ticagrelor monotherapy after Ultimaster stent implantation	POCE at 1 y	February 2030
SORT OUT XII DAPT	Open-label NCT06718179	3,150	ACS	1-mo DAPT with prasugrel followed by prasugrel monotherapy (11 mo)	12-mo prasugrel-based DAPT	BARC 2, 3, or 5 bleeding and MACCE at 1 y	December 2030
With concomitant indication for OAC							
MATRIX-2	Open-label NCT05955365	3,010	ACS or CCS undergoing PCI	P2Y ₁₂ inhibitor monotherapy up to 1 mo (omitting clopidogrel nonresponders) followed by OAC monotherapy	Triple therapy with DOAC for at least 12 mo, aspirin for up to 1 mo after PCI and a P2Y ₁₂ inhibitor at investigator's discretion for a minimum of 6 mo and up to 12 mo after PCI	MACCE and major or clinically relevant nonmajor bleeding according to ISTH criteria at 1 y	December 2026
WOEST-3	Open-label NCT04436978	2,000	ACS or CCS undergoing PCI	30-d DAPT with aspirin and a P2Y ₁₂ inhibitor (at discretion of physician) followed by edoxaban and a P2Y ₁₂ inhibitor	Triple therapy with edoxaban, a P2Y ₁₂ inhibitor, and aspirin up to 30 d followed by dual therapy with edoxaban and P2Y ₁₂ inhibitor	MACCE and major or clinically relevant nonmajor bleeding according to ISTH criteria at 6 wk	December 2027

ACS = acute coronary syndrome; BARC = Bleeding Academic Research Consortium; CCS = chronic coronary syndrome; CKD = chronic kidney disease; DAPT = dual antiplatelet therapy; DES = drug-eluting stent; HBR = high bleeding risk; ISTH = International Society of Thrombosis and Haemostasis; IVUS = intravenous ultrasound; MACE = major adverse cardiovascular events; MACCE = major adverse cardiac and cerebrovascular events; NACE = net adverse clinical events; OAC = oral anticoagulant; PCI = percutaneous coronary intervention; POCE = patient-oriented composite endpoint; PPPI = primary percutaneous coronary intervention; ST = stent thrombosis; STEMI = ST-segment elevation myocardial infarction.

ischemic efficacy; this strategy is now supported by strong evidence across diverse geographic settings, with timing of de-escalation ranging from 1 month in patients at low thrombotic risk and after 3 months in the broader ACS population. Finally, situations in which aspirin withdrawal strategies should be avoided in patients with CAD include: 1) before 30 days after ACS/PCI; 2) early (<3 months after ACS) aspirin withdrawal when followed by unguided clopidogrel monotherapy; and 3) very early (<7 days) aspirin withdrawal in ACS/PCI patients at high thrombotic risk who also have a concomitant indication for OAC.

GAPS IN EVIDENCE AND LIMITATIONS OF THE CURRENT LITERATURE

Several important gaps in evidence remain regarding aspirin discontinuation strategies for secondary prevention in CAD (**Central Illustration**). First, comparative data on different de-escalation

strategies—including aspirin withdrawal stratified by the type and dose of P2Y₁₂ inhibitor, pharmacodynamic response, or geographic region of enrollment—remain scarce. Emerging data suggest that prasugrel may be associated with more favorable outcomes than ticagrelor^{136,137}; however, though prasugrel is associated with similar levels of P2Y₁₂ inhibitory effects compared with ticagrelor, these pharmacodynamic findings cannot be extrapolated to practice recommendations, because the level of clinical trial evidence for prasugrel monotherapy after a brief period of DAPT is limited, and no trials have directly compared prasugrel vs ticagrelor monotherapy. In addition, local availability, cost considerations, and clinician familiarity may influence implementation. Second, in an era of increasing emphasis on precision medicine, interindividual variability in response to antiplatelet therapy cannot be overlooked.³⁷ This is particularly relevant for clopidogrel, the most widely used P2Y₁₂ inhibitor, for which nonresponsiveness

affects ~30% of treated patients, depending on ancestry, and is associated with a well-established increase in thrombotic risk.²⁹ This concern may be particularly relevant when clopidogrel is used as monotherapy or in combination with an OAC, because nonresponders may derive minimal antiplatelet benefit in the absence of background aspirin therapy; however, clinical evidence supporting this hypothesis remains limited.²⁹ At the same time, in patients identified as responders, clopidogrel provides a more favorable safety profile (ie, less bleeding) with no apparent loss of ischemic efficacy compared with prasugrel or ticagrelor.^{55,57,58} This concept is central to optimizing the balance between ischemic and bleeding risk, particularly in selected patient subgroups, such as those at heightened risk for both bleeding and ischemic events, known as bi-risk (eg, older patients with ACS), or patients with CCS and high ischemic risk defined by clinical or procedural characteristics.²⁹ This approach is consistent with recent international consensus documents and position papers that support the use of PFT or GT to guide antiplatelet therapy, particularly in selected patients (eg, bi-risk) with CAD.^{29,34} Because bleeding and ischemic risks frequently overlap, guiding antiplatelet therapy solely based on categorizing patients as having HBR or high thrombotic risk may be of limited practical value for a substantial proportion of patients encountered in routine clinical practice. This challenge is further compounded by real-world data showing limited implementation of formal risk stratification tools.¹³⁴ Third, it is unknown whether monotherapy with low-dose prasugrel (5 mg daily) or ticagrelor (60 mg twice daily), initiated after a short course of DAPT (1–3 months) or at the time of ACS/PCI in patients identified as clopidogrel nonresponders, improves outcomes compared with continuation of a standard dose of prasugrel (10 mg daily) or ticagrelor (90 mg twice daily) for up to 12 months. This question is particularly relevant given the well-established temporal decline in thrombotic risk after ACS/PCI contrasted with a relatively stable bleeding risk over time. As a result, prolonged exposure to prasugrel or ticagrelor monotherapy at a standard dose may increase bleeding without a commensurate reduction in ischemic events in selected patients. Moreover, the use of this standard dose regimen has not been extensively studied beyond 12 months, and the only reduced-dose regimen—60 mg ticagrelor twice daily—has been evaluated exclusively in combination with aspirin and is indicated for selected patients (ie, increased ischemic and low bleeding risks) with CCS.^{61,62,138} Fourth, the perioperative management

of patients on P2Y₁₂ inhibitor monotherapy who require cardiac or noncardiac surgery remains challenging, and standardized management protocols are currently lacking. Fifth, the extent to which the safety and efficacy of aspirin withdrawal are modified by patient- and procedure-related factors such as sex, ethnicity, PCI complexity, PCI techniques (eg, stent vs drug-coated balloon), use of intraprocedural imaging to optimize PCI, age, body mass index, chronic kidney disease, or diabetes, as well as by different aspirin formulations (ie, chewable vs enteric-coated), requires further investigation. Sixth, it remains uncertain whether the safety and efficacy of aspirin withdrawal in patients with a concomitant indication for OAC extend to those receiving low-intensity anticoagulation (eg, for previous venous thromboembolism). Finally, it remains unclear whether the use of recently available therapies targeting residual cardiovascular risk could enable further de-escalation of antiplatelet therapy in contemporary CAD while maintaining ischemic protection and reducing bleeding risk.^{3,65}

FUTURE PERSPECTIVES AND CONCLUSIONS

Aspirin withdrawal represents a major paradigm shift in the secondary prevention of CAD, supported by a growing body of randomized evidence demonstrating meaningful reductions in bleeding without an apparent trade-off in ischemic protection in many patients. Although robust evidence now supports aspirin withdrawal when implemented 1 to 3 months after the index event in patients treated with potent P2Y₁₂ inhibitors, some knowledge gaps remain that limit its universal adoption. These include uncertainties regarding optimal patient selection, choice, and duration of P2Y₁₂ inhibitor monotherapy, the role of PFT and GT to guide therapy, and the generalizability of trial results across diverse populations. Future trials (Table 4) and guideline updates should focus on refining personalized risk-adapted aspirin withdrawal that balances ischemic and bleeding risks across the full spectrum of patients encountered in contemporary clinical practice.

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